

"For body under constant ~~acceler~~ force we measure the position of the body for the following time points ~~2~~ $t = (-2, -1, 1, 2)$ seconds as $x = (3, 24, 26, 17)$ meters, respectively.

* Find the ~~best approximation~~ for initial position, ~~initial~~ speed and acceleration $x(0)$, $v(0)$, a .

$$x(t) = x_0 + v_0 t + \frac{a}{2} t^2 \rightarrow x(t) = 30 + 3t - 5t^2$$

$$\begin{aligned} 3 &= x_0 + 2v_0 + 4\frac{a}{2} \\ 24 &= x_0 - v_0 + \frac{a}{2} \\ 26 &= x_0 + v_0 + \frac{a}{2} \\ 17 &= x_0 + 2v_0 + 4\frac{a}{2} \end{aligned} \Rightarrow \begin{matrix} x_0 \\ v_1 \\ v_2 \\ v_3 \end{matrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{matrix} v_0 \\ v_1 \\ v_2 \\ v_3 \end{matrix} \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} + \frac{a}{2} \begin{bmatrix} 4 \\ 1 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 24 \\ 26 \\ 17 \end{bmatrix}$$

(i) Is vector x in the span of v_1, v_2, v_3 i.e.

i.e. do we have a solution for LS defined by the coefficient matrix

$$\begin{bmatrix} 1 & -2 & 4 & 3 \\ 1 & -1 & 1 & 24 \\ 1 & 1 & 1 & 26 \\ 1 & 2 & 4 & 17 \end{bmatrix} \xrightarrow{\substack{R_3 - R_1 \rightarrow R_2 \\ R_4 - R_1 \rightarrow R_3}} \begin{bmatrix} 1 & -2 & 4 & 3 \\ 0 & 1 & -3 & 21 \\ 0 & 3 & -3 & 23 \\ 0 & 4 & 0 & 14 \end{bmatrix} \xrightarrow{\substack{R_3 - 3R_2 \rightarrow R_3 \\ R_4 - 4R_2 \rightarrow R_4}} \begin{bmatrix} 1 & -2 & 4 & 3 \\ 0 & 1 & -3 & 21 \\ 0 & 0 & 6 & -40 \\ 0 & 0 & 12 & -70 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -2 & 4 & 3 \\ 0 & 1 & -3 & 21 \\ 0 & 0 & 6 & -40 \\ 0 & 0 & 0 & 10 \end{bmatrix} \rightarrow \text{inconsistent system, we do not stop here, find the best approximation.}$$

(1) find orthonormal basis for $\text{span}\{v_1, v_2, v_3\}$

(i) $\hat{e}_1 = \frac{v_1}{\|v_1\|} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, (ii) $a_{2p} = (\hat{e}_1, v_2) \hat{e}_1 = 0 \cdot \hat{e}_1 = 0$

$$\hat{e}_2 = \frac{v_2 - a_{2p} \hat{e}_1}{\|v_2 - a_{2p} \hat{e}_1\|} = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{10}} \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$$

(iii) $v_{3p} = (\hat{e}_1, v_3) \hat{e}_1 + (\hat{e}_2, v_3) \hat{e}_2 = \left(\frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 1 \\ 1 \\ 4 \end{bmatrix} \right) \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \frac{5}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $\hat{e}_3 = \frac{v_3 - v_{3p}}{\|v_3 - v_{3p}\|} = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$

$$\text{span}\{v_1, v_2, v_3\} = \text{span}\{\hat{e}_1, \hat{e}_2, \hat{e}_3\}$$

(2) Project x onto $\text{span}\{\hat{e}_1, \hat{e}_2, \hat{e}_3\}$ to find $\tilde{x}_p$

$$\begin{aligned} \tilde{x}_p &= (\hat{e}_1, x) \hat{e}_1 + (\hat{e}_2, x) \hat{e}_2 + (\hat{e}_3, x) \hat{e}_3 = \left(\frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 24 \\ 26 \\ 17 \end{bmatrix} \right) \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \left(\frac{1}{\sqrt{10}} \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 24 \\ 26 \\ 17 \end{bmatrix} \right) \frac{1}{\sqrt{10}} \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} + \left(\frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 24 \\ 26 \\ 17 \end{bmatrix} \right) \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \\ &= \frac{35}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{3}{\sqrt{10}} \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} + \frac{15}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 22 \\ 28 \\ 16 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 4 & 4 \\ 0 & 1 & -3 & 18 \\ 0 & 3 & -3 & 24 \\ 0 & 4 & 0 & 12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 4 & 4 \\ 0 & 1 & -3 & 18 \\ 0 & 0 & 6 & -30 \\ 0 & 0 & 12 & -60 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 4 & 4 \\ 0 & 1 & -3 & 18 \\ 0 & 0 & 6 & -30 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$6 \cdot \frac{a}{2} = -30 \quad a = -10 \quad v_0 = 18 + 3 \frac{a}{2} = 18 - 15 = 3 \quad x_0 = 44 - 4 \frac{a}{2} + 2v_0 = 4 + 20 + 6 = 30$$